Description

Illustration



Figure caption

1. Pile of dry grey portland cement. 2. Cement (mortar) inside a cement mixer. 3. Polished concrete made from Portland cement. Images used under license from Shutterstock.com

The material

Cement is arguably one of the four man-made structural materials that have most changed the world, to be ranked with bronze, iron (steel) and glass. Roman cement, still intact after 2000 years, enabled some of their greatest constructions. Today cement production exceeds 1.5 billion tonnes per year.
Cement is a calcined, powdered ceramic that, with the addition of water, can be molded or poured and then sets into a solid mass, bonding well to many surfaces. There are many different cements; the one most widely used construction is Portland cement, so called because the developers sought to promote it as resembling Portland stone, a layered limestone much valued for floors and pavings (the resemblance is a poor one). It is made by calcining a mixture of chalk or limestone and clay, followed by grinding to a fine powder. Mixed with water, it hydrates and sets to a hard gray solid. This process is referred to as curing, releasing heat and resulting in a small shrinkage of the concrete. Portland cement can be used neat, but rarely is in practice - being more usually used as a binder with aggregates to make mortar and concrete. As a constituent of concrete, cement is the primary binding agent bringing together aggregates, sand, water and steel reinforcing to make the most widely used structural material in the world today.
The properties of cement are highly dependent on its composition, particularly the water:cement ratio. The quoted properties are for cement with a water:cement ratio by weight of 0.65.

Compositional summary

63% CaO, 21% SiO2, 6% Al2O3 + additions
water:cement ratio by weight = 0.65

General properties

Density	1.8e3	-	2.2e3	kg/m^3
Price	* 0.0755	-	0.0906	GBP/kg
Date first used	-200			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	*	40.2	-	41.6	GPa
Shear modulus	*	16.5	-	17	GPa
Bulk modulus	*	23.9	-	24.8	GPa
Bending modulus	*	40.2	-	41.6	GPa
Poisson's ratio		0.2	-	0.24	
Yield strength (elastic limit)	*	1.9	-	2.1	MPa
Tensile strength	*	1.9	-	2.1	MPa
Compressive strength	*	18.7	-	20.7	MPa
Bending strength		1.9	-	2.1	MPa
Elongation	*	0	-	0.01	% strain
Hardness - Vickers	*	5.6	-	6.2	HV
Fatigue strength at 10^7 cycles	*	0.9	-	1.41	MPa
Fracture toughness		0.35	-	0.45	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		1.11	-	1.25	m.°C/W
Thermal expansion coefficient		15	-	22	µstrain/°C
Specific heat capacity	*	813	-	867	J/kg.°C
Melting point		927	-	1.2e3	°C
Maximum service temperature	*	627	-	857	°C
Minimum service temperature		-160	-	-150	°C
Flammability		Non-flammable			
Emissivity		0.54			

Hygro-thermal properties

Water vapor permeability		9.76e-12	-	2.6e-11	kg.m/s.m^2.Pa
Frost resistance		Poor			

Electrical properties

Electrical conductor or insulator?	Poor insulator			
Electrical resistivity	2.06e11	-	2.06e12	µohm.cm
Dielectric constant (relative permittivity)	* 8	-	12	
Dissipation factor (dielectric loss tangent)	* 0.001	-	0.01	
Dielectric strength (dielectric breakdown)	1.44	-	1.59	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Poor
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Castability	4	-	5
Moldability	4	-	5

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Unacceptable
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Limited use
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Unacceptable
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Geo-economic data for principal component

Annual world production, principal component	4.18e9			tonne/yr

Primary material production: energy and climate change

Climate change (CO2-eq) (virgin grade production)	* 0.853	-	0.941	kg/kg
Embodied energy (virgin grade production)	* 4.66	-	5.13	MJ/kg
Water Usage	* 35.1	-	38.8	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 17.6	-	19.1	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.15	-	1.25	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	1.14	-	1.26	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Portland cement, used as a constituent of grouts, mortars and concrete, can be used for a wide variety of applications.

ASTM C 150 (the US standard) specifies eight separate types of Portland cement that can be applied to almost any construction need; Type I is Normal cement, Type IA Normal, air entraining, Type II Moderate resistance to sulfate attack, Type IIA Moderate resistance, air entraining, Type III High early strength, Type IIIA High early strength, air entraining, Type IV Low heat of hydration, Type V High resistance to sulfate attack. For most construction applications, Type I is used. Types II and V are used in situations in which the material will come in contact with high sulfate-containing water. Type IV is reserved for very large, massive structures in which the heat of hydration needs to be carefully controlled.

EN 197-1 (the European standard) specifies 27 different types of common cements available in Europe, based on CEM 1 (Portland cement clinker), but also supplied pre-blended in varying proportions with one or more other powders. These other powders are called Alternative Cementing Materials (ACMs) and principally include fly ash and natural pozzolana, ground granulated blastfurnace slag, and silica fume. Which ACMs are used and in which proportion is classified by the standards CEM II, III IV & V and sub-categories. Use of the different cements in concrete is permitted through EN 206-1, although the specific benefits are left to National guidance applicable to the place of use. In the UK, the National recommendations are given in BS 8500.

As recognised in BS 8500-1, Portland cement with ACMs provides a range of benefits over 100% CEM I, including increased sulfate and chloride ion resistance and reduced heat of hydration for pours of minimum dimension above 0.5m. BS 8500-2 also permits "combination" cements to be made by adding ACMs directly into the concrete mix at the concrete batch plant, rather than being pre-blended with CEM I (as required by EN 197-1).

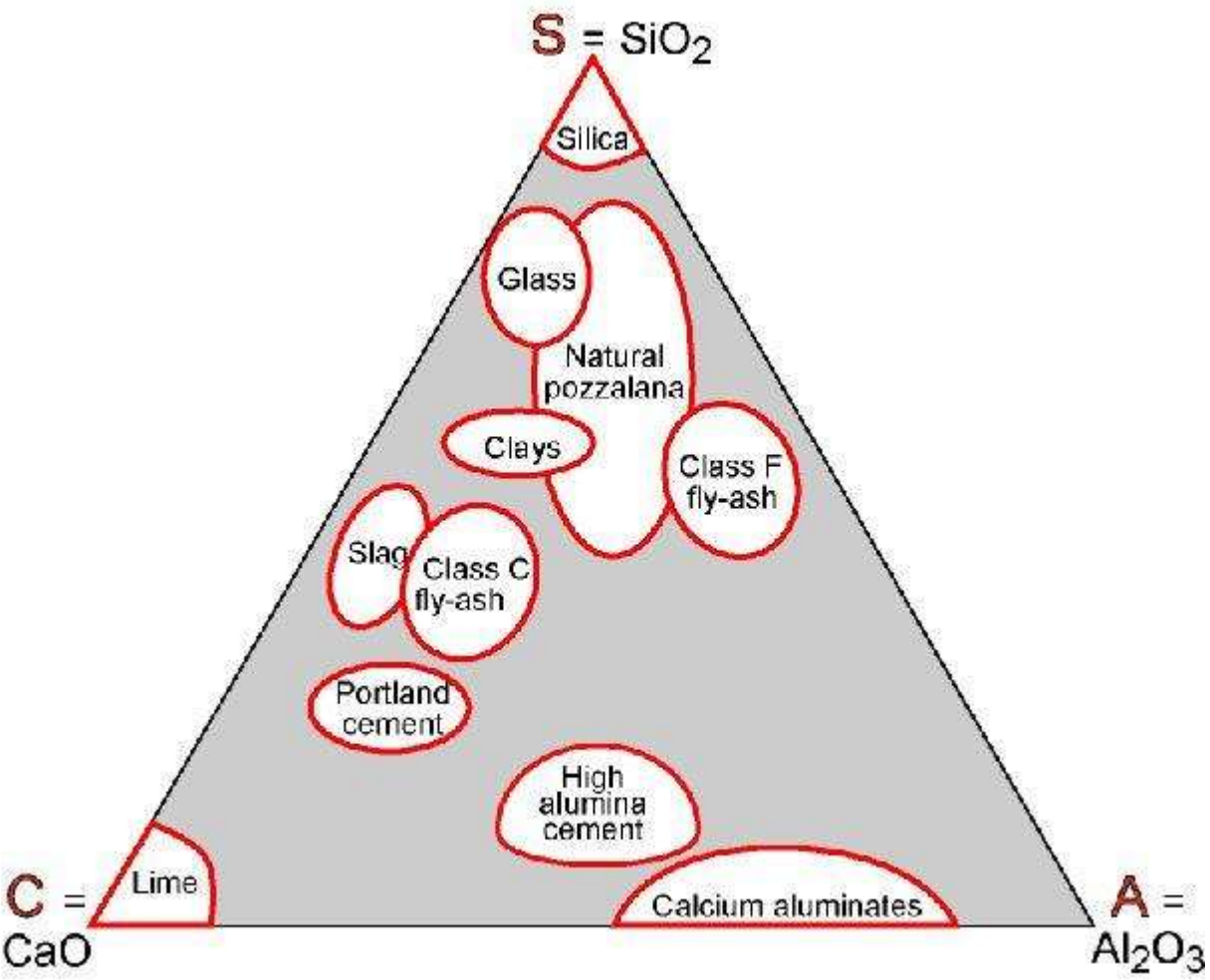
Technical notes

Cements, clays and certain glasses are based on 3 main ingredients: silica (SiO_2), alumina (Al_2O_3) and lime (CaO), sometimes with a fourth, iron oxide (Fe_2O_3). In cement chemistry these main ingredients are referred to a S, A, C and F, thus the main constituents are reported as C_3S (meaning $3(CaO).SiO_2$), C_3A (meaning $3(CaO).Al_2O_3$) and so on.

Portland cement is made by calcining (heating at 1500C) a mixture of chalk and clay. The triangular diagram shows the composition of Portland cement.
High alumina cement is also shown. It hardens more rapidly than Portland cement, cutting construction costs, but its incorrect use in reinforced concrete structural applications has led to failures.

The physical and chemical properties of a cement are highly dependent on its composition, whereas the pore structure of the hydrated cement and its permeability to gas and water in the hardened product (grout, mortar or concrete), depends principally on the water/cement ratio. The quoted properties are for cement with a water/cement ratio by weight of 0.65.

Technical notes images



Typical uses

Used to make grouts, mortars and concrete for general civil engineering and building construction.
Portland cement (CEM I) is commonly used where there are low concentrations of sulfate ions in soil or groundwater.

Most cement supplied in Europe today is CEM II, containing partial replacement of cement with by-products, so reducing the carbon footprint.

High sulfate levels in soils and severe chloride ion exposures (seawater, de-icing salts) benefit from cement compositions containing ACMs (e.g. CEMIII or CEM IV) or additional protection (e.g surface protection systems).

Tradenames

Portland cement (CEM1), Fly ash, Ground granulated blastfurnace cement, Silica fume

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
ASTM C1709: Standard Guide for Evaluation of Alternative Supplementary Cementitious Materials (ASCM) for Use in Concrete
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity
BS 8500: Concrete. Complementary British Standard to BS EN 206
BS 8500-1 - specifying concrete
BS 8500-2 - constituent materials

Further reading

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Links

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